

8 The equations of three planes are

$$2x + y + 3z = 3,$$

$$3x - y - 2z = 2,$$

$$-4x + 3y + 7z = k,$$

where k is a constant.

(a) By considering a suitable determinant, show that the planes do **not** meet at a single point. **[2]**

(b) Given that the planes form a sheaf, determine the value of k . **[4]**